

**ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)**  
**ACADEX PRACTICE EXAMINATION**  
**A-LEVEL · FURTHER MATHEMATICS**

**FURTHER MATHEMATICS 9187/1**

**PAPER 1 November 2023**

**3 hours**

**PAPER 1 — SHORT-ANSWER — CALCULATORS MUST NOT BE USED**

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2023 9187/1.

**Additional materials:** A calculator may be used unless a question forbids it.

**INSTRUCTIONS:** Answer all questions. Full marks are not given for answers only — show method.

- 1.**  $z = 3 + 4i$ . Find  $|z|$ ,  $\arg z$ , and  $\bar{z}$ . Express  $1/z$  in the form  $a+bi$ . **[10]**
- 2.**  $A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ . Find  $\det A$ ,  $A^{-1}$ , and  $A^2$ . **[10]**
- 3.** Prove by induction that  $1 + 2 + \dots + n = n(n+1)/2$  for  $n \geq 1$ . **[10]**
- 4.** Write  $1 + i$  in modulus-argument form and find  $(1+i)^4$  using De Moivre. **[8]**
- 5.** Use identities to simplify  $\sin 3\theta$  in terms of  $\sin \theta$  (or state  $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$ ). **[8]**
- 6.** Show that  $(i+j) \times (i+2j+k) = i - j + k$  or compute the cross product of  $a=(1,1,0)$ ,  $b=(1,2,1)$ . **[8]**
- 7.** Standard Further Pure question on Maclaurin series. Show full working. **[8]**
- 8.** Standard Further Pure question on Reduction formula sketch. Show full working. **[8]**
- 9.** Standard Further Pure question on Eigenvalues of  $(2,1;1,2)$ . Show full working. **[8]**
- 10.** Standard Further Pure question on Second order DE  $y''+y=0$ . Show full working. **[8]**
- 11.** Standard Further Pure question on Further partial fractions. Show full working. **[8]**

**12.** Standard Further Pure question on Argand diagram loci. Show full working. **[8]**

**END OF QUESTION PAPER**

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).